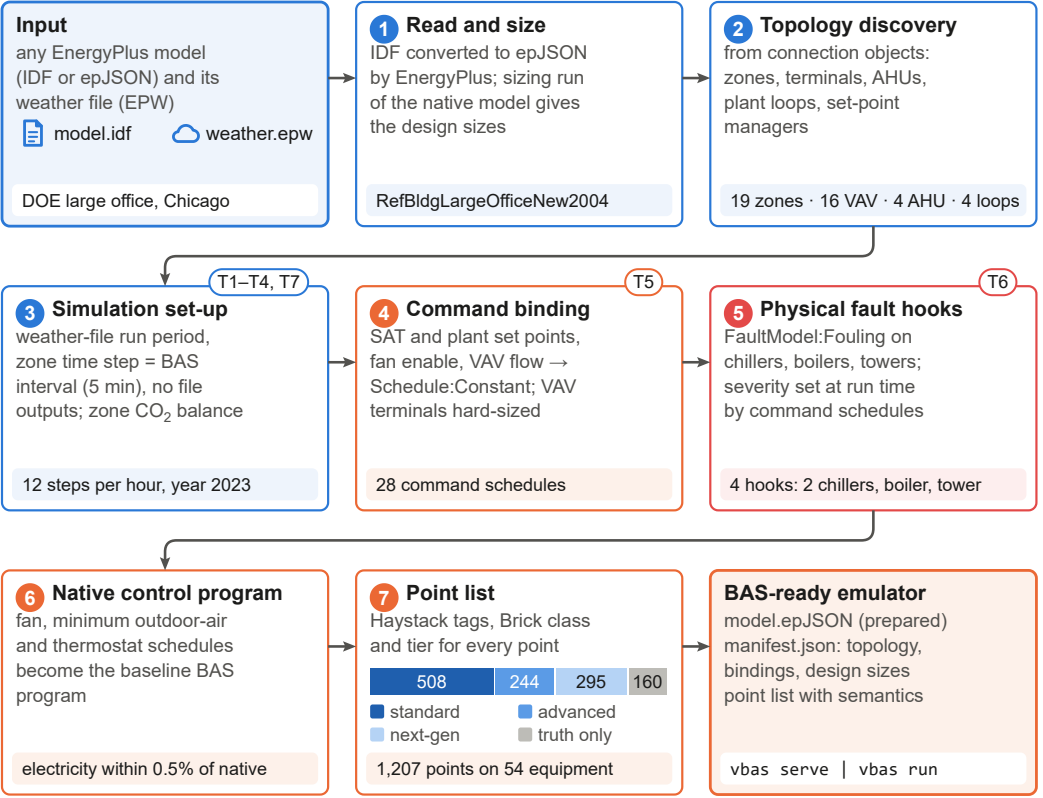

Input
any EnergyPlus model (IDF or epJSON) and its weather file (EPW)
model.idf
weather.epw
DOE large office, Chicago
1 Read and size
IDF converted to epJSON by EnergyPlus; sizing run of the native model gives the design sizes
RefBldgLargeOfficeNew2004
2 Topology discovery
from connection objects: zones, terminals, AHUs, plant loops, set-point managers
19 zones · 16 VAV · 4 AHU · 4 loops
T1–T4, T7
3 Simulation set-up
weather-file run period, zone time step = BAS interval (5 min), no file outputs; zone CO2 balance
12 steps per hour, year 2023
T5
4 Command binding
SAT and plant set points, fan enable, VAV flow → Schedule:Constant; VAV terminals hard-sized
28 command schedules
T6
5 Physical fault hooks
FaultModel:Fouling on chillers, boilers, towers; severity set at run time by command schedules
4 hooks: 2 chillers, boiler, tower
6 Native control program
fan, minimum outdoor-air and thermostat schedules become the baseline BAS program
electricity within 0.5% of native
7 Point list
Haystack tags, Brick class and tier for every point
508
244
295
160
standard
advanced
next-gen
truth only
1,207 points on 54 equipment
BAS-ready emulator
model.epJSON (prepared)
manifest.json: topology, bindings, design sizes
point list with semantics
vbas serve | vbas run